

20 LNG Questions for RAG Agent Testing

A balanced test set covering LNG fundamentals, infrastructure, contracts, pricing, shipping, risk, emissions, and project economics. These questions are designed to test retrieval quality, grounding, synthesis, and citation behavior.

- 1 What are the main stages of the LNG value chain, from natural gas production to delivery to the end customer?
- 2 How does a liquefaction plant convert natural gas into LNG, and why is the gas cooled to approximately -162°C ?
- 3 What factors determine the nameplate capacity and actual utilization rate of an LNG export terminal?
- 4 What is the difference between baseload LNG, peak-shaving LNG, and small-scale LNG projects?
- 5 How do Henry Hub, TTF, and JKM pricing benchmarks differ, and where is each benchmark most relevant?
- 6 What are the main pricing structures used in long-term LNG sales and purchase agreements, including oil-indexed and gas-indexed contracts?
- 7 What is the difference between FOB and DES LNG contracts, and how do those terms affect shipping responsibility and commercial risk?
- 8 What factors influence LNG shipping costs, including voyage distance, vessel size, boil-off gas, canal fees, and charter rates?
- 9 How does boil-off gas affect LNG storage and transportation, and how is it typically managed on modern LNG carriers?
- 10 What is the role of regasification terminals, and how do onshore terminals differ from floating storage and regasification units (FSRUs)?
- 11 How do seasonal demand patterns influence LNG flows between Asia, Europe, and the Atlantic Basin?
- 12 What are the principal drivers of LNG demand in power generation, industry, residential heating, and transportation?
- 13 How can LNG projects reduce lifecycle greenhouse-gas emissions, including methane leakage, liquefaction energy use, shipping emissions, and regasification?
- 14 What are the main technical and commercial risks faced by a new LNG export project before final investment decision (FID)?
- 15 How do feedgas availability, pipeline connectivity, and upstream gas supply affect the economics and reliability of an LNG project?
- 16 What role do long-term offtake agreements play in financing LNG projects, and why are creditworthy buyers important?
- 17 How do geopolitical disruptions, sanctions, canal closures, or regional conflicts affect LNG trade routes and market prices?

- 18 What is the difference between brownfield and greenfield LNG development, and what are the main advantages and disadvantages of each?
- 19 How are LNG terminals and projects evaluated for energy security, data sovereignty, cybersecurity, and operational resilience?
- 20 If you were comparing two LNG projects, which technical, financial, contractual, environmental, and market metrics would you use to determine their relative competitiveness?

Tip for testing: Ask the agent to cite the source document and page for each answer, and include a clear "not found in the source" response when the corpus does not support the answer.